EXP-005 — Re-Chunking Results

Testing whether chunk granularity is the retrieval bottleneck the previous failure report said it was.

VERDICT

The chunk-granularity hypothesis is not supported. Bounding chunk size — the intervention the hypothesis actually names — rescued **zero** of the 20 questions, despite cutting the corpus maximum from 16,096 characters to 1,999.

The one case that motivated the entire hypothesis is still unretrievable after its evidence share of its chunk improved 2.9×. Its real cause is vocabulary mismatch, which no chunker can fix.

0 rescued

Questions recovered by bounded chunking (EXP-005A), the isolated granularity intervention

8 up / 9 down

Span ranks that improved vs worsened under bounded chunking — a coin flip, and no span became newly reachable

Controls

Held constant: the same 202 document versions, the same 20 questions and 22 evidence spans, the same anchors `(version_id, section_path, char_start, char_end)`, the same BM25 implementation and parameters, the same `k=10`, the same scoring code. No reranker, no query rewriting, no BM25 retuning, no larger k.

Storing a second chunking of the same versions required a chunk-set dimension in the schema. After migrating, EXP-000 was re-run against the control set and its results file is **byte-identical** to the committed one.

Chunk distribution — the intervention was real

Chunker	chunks	mean	median	p90	p99	max	>2000	> own limit
v1 control	14,209	1,097	508	3,466	3,500	**16,096**	3,069	none
v2 bounded	20,526	757	917	1,193	1,879	**1,999**	0	0
v3 technical	20,516	764	907	1,192	1,837	**1,999**	0	0

Mean chunk length fell 31%, p90 fell from 3,466 to 1,193, and 3,069 chunks over 2,000 characters became none. Evidence mapping was validated: 22/22 spans still map to a chunk with the same section path in all three sets, with zero offset mismatches.

Results

Configuration	macro recall	fully recalled	spans	doc recall	rescued / regressed
EXP-000 control	0.475	9/20	10/22	0.825	—

Configuration	macro recall	fully recalled	spans	doc recall	rescued / regressed
EXP-005A bounded	0.500	9/20	11/22	0.825	0 / 0
EXP-005B technical	0.650	12/20	14/22	0.900	3 / 0

Span-level rank movement (all 22 spans)

Configuration	improved	worsened	unchanged	newly reachable
EXP-005A bounded	8	**9**	5	**0**
EXP-005B technical	10	6	6	**0**

Per-question movement

Case	Category	ctrl	V2	V3	rank ctrl	rank V2	rank V3
AN-001	exact_lookup	0.00	0.00	0.00	19	29	32
AN-002	exact_lookup	0.00	0.00	0.00	27	172	171
AN-003	exact_lookup	0.00	0.00	0.00	—	—	—
AN-004	exact_lookup	0.00	0.00	1.00	12	11	10
AN-005	exact_lookup	1.00	1.00	1.00	4	3	1
AN-006	exact_lookup	0.00	0.00	0.00	29	36	32
AN-007	exact_lookup	0.00	0.00	0.00	18	34	41
AN-008	normal	0.00	0.00	1.00	12	11	9
AN-009	version_conflict	1.00	1.00	1.00	1	1	1
AN-010	version_conflict	0.00	0.00	1.00	49	62	1
AN-011	ambiguous	0.00	0.00	0.00	54	153	139
AN-012	multi_hop	0.00	0.50	0.50	47,117	6,223	6,248
OA-001	exact_lookup	1.00	1.00	1.00	4	7	4
OA-002	exact_lookup	1.00	1.00	1.00	2	1	1
OA-003	exact_lookup	1.00	1.00	1.00	6	4	4
OA-004	exact_lookup	1.00	1.00	1.00	5	4	4
OA-005	exact_lookup	1.00	1.00	1.00	3	3	1
OA-006	multi_hop	0.50	0.50	0.50	1,29	1,25	1,18
OA-007	normal	1.00	1.00	1.00	1	2	1

Case	Category	ctrl	V2	V3	rank ctrl	rank V2	rank V3
OA-008	exact_lookup	1.00	1.00	1.00	1	1	1

The three cases that explain the result

Major rescue — AN-010, rank 49 → 1 (V3 only)
The evidence is a row in a model-status table. Control: a 2,317-character chunk, rank 49. V2 shrank it to 1,871 and the rank got **worse** (62), because splitting removed the surrounding prose carrying `state` and `model`. V3 emitted a 415-character row group with a context header repeating `[Model status] | API model name | Current state | …` — rank 1. The mechanism is the header, not the size, and V2 proves it.

Regression — AN-002, rank 27 → 172 (V2)
Splitting a 3,327-character chunk to 1,121 cut query-term coverage from **12 of 13** to **7 of 13**. The control's oversized chunk was accidentally helping, because BM25 rewards co-occurrence and that chunk concatenated the whole HTTP-error list. AN-007 (18→41) and AN-011 (54→139) fail the same way.

Persistent failure — AN-003, the case that motivated the hypothesis
The answer is a 57-character sentence. Its share of its chunk improved from 1.65% to 4.79% — a 2.9× improvement on exactly the quantity that was diagnosed — and it is **still not retrieved at depth 300** in any configuration. Every low-df term in the question is absent from the chunk holding the answer: `contain` (df 111), `Batches` (df 286), `requests` (df 1,218 — only the singular appears). This is vocabulary mismatch.

Answers to the pre-registered questions

A. Did span recall move toward the 0.818 document ceiling?
Partially, and not by the mechanism under test. V2, which isolates granularity: 0.475 → 0.500, entirely one half-case.

B. Did document recall stay stable?
For V2 yes (0.825). For V3 no — it rose to 0.900, because context headers add section-path terms to the index. V3 is therefore not a pure chunking intervention.

C. Did oversized answer-containing units become retrievable?
No. Zero spans moved from unreachable to reachable in either configuration.

D. Did smaller chunks introduce fragmentation regressions?
Yes. Under V2, 9 of 22 spans ranked worse against 8 better.

E. Does V3 justify its complexity?
On the numbers yes (+3 rescued, 0 regressed). But the justification is for contextual enrichment, which V3 bundles — not for its parameter-entry logic, which produced only 60 chunks across 202 documents and rescued nothing.

Limitations

1. **V2 is not a single-variable ablation** — it changes both the grouping target and hard-limit enforcement. Since its result is null, the confound does not rescue the hypothesis, but a stricter design would vary one at a time.
2. **V3 confounds granularity with enrichment.** Separating them is the obvious follow-up.

3. **n = 20 is development scale.** Two of the three V3 rescues turned on rank movements of 1–3 positions across the k boundary. Nothing here is statistically significant.
4. **EXP-NULL still has not run** — no generation credential and the provider host is egress-blocked. Retrieval lift over the model's own knowledge remains unknown.
5. **Dense retrieval remains unmeasured in any real sense.** Earlier dense numbers used an offline TF-IDF+SVD substitute. Nothing supports "BM25 beats dense retrieval" — only that it beat that substitute on this corpus. EXP-005 is BM25-only.
6. **Corpus skew persists:** 139 Anthropic documents to 63 OpenAI. All three V3 rescues are Anthropic-side; the OpenAI cases are unchanged across all configurations.

What the data justifies next

The bottleneck is **lexical matching**, not chunk size.

1. **A real pretrained embedding model** — highest value. AN-003 fails on pure vocabulary mismatch, the canonical case for semantic retrieval. Blocked only on network egress.
2. **Separate enrichment from granularity:** run V2 plus context headers alone. If it reproduces most of V3's +3, adopt enrichment and drop the parameter-entry machinery.
3. **Revisit the unstemmed `simple` configuration** as a third ranked list, keeping identifier precision while recovering plural/singular matches.

Not justified: a reranker (cannot rescue evidence absent from the top 300), a larger k (inflates the metric without improving retrieval), or a confidence threshold (no calibration failure was observed). **V3 is not frozen as the new baseline**, because its gain is attributable to a mechanism this experiment did not isolate.

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